Alberto Nieto

PRINCIPAL ENGINEER | AI, LLM EVALUATION & AGENTS

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Phoenix, Arizona

PROFESSIONAL SUMMARY

Principal Engineer with 25+ years in financial services, from banking operations to principal-level AI engineering. Builds agentic AI, LLM evaluation and decision models, document and video pipelines, and the cloud infrastructure they run on (GCP, OpenShift/Kubernetes). Outside work, builds open source tools that test what models actually do, then publishes the evidence: 8 Python packages on PyPI. Co-inventor on three pending patent applications. Bilingual in English and Spanish.

WORK EXPERIENCE

Principal Engineer (Executive Director)

Wells Fargo

Builds AI solutions for line-of-business needs, plus the documentation and frameworks that speed developer onboarding and production rollouts.

Phoenix, Arizona

Jul 2025 - Present

- Created AI agent framework to automate line-of-business operations, adopted across multiple teams
- Proposed and built the first fine-tuning of Gemma 3 and Gemma 4 on internal policy documents (October 2026), toward a decision model whose answers are read as next-token probabilities: graphed the corpus into entities and relationships to build the training set, refined it from training logs and sample outputs, and held out data for evaluation. Supported a demo to AI leadership.
- Building decision models for collections and customer-service next-best-action
- Built video processing pipeline handling thousands of videos daily into standard operating procedures for knowledge tree persistence
- Implemented evaluation and evolution frameworks to score and evolve agent definitions for data pipeline tasks (open-sourced as gepa-adk)
- Deploys production workloads to OpenShift (OCP), with Kafka for messaging and Grafana for monitoring
- Built AI workflows whose volume pushed the bank's internal enterprise LLM API to its capacity limits
- Established patterns ensuring non-functional requirements are satisfied with AI deployments
- Audited enterprise API clients for Apigee AI provisioning to enable multi-modal capabilities and proper scaling

Senior Lead Analytics Consultant (Executive Director)

Wells Fargo

Individual contributor and the team's go-to engineer for GenAI and agentic AI frameworks. Focused on proof of concepts and production-ready AI implementations.

Phoenix, Arizona

Jun 2022 - Jul 2025

- Architected scalable OCR pipeline processing 500K+ documents using Gemini, with results stored in Teradata and SQLite for distributed processing
- Developed intelligent agents for database querying, KPI summarization, SQL validation, and structured QA using Python and Google ADK
- Built reusable data pipelines for document ingestion, OCR, and validation using Selenium, HTTPX, and Docling with Gradio/Streamlit frontends
- Designed modular, scalable architecture patterns reused across multiple teams and projects
- Implemented git-based dev workflow with peer reviews, pull requests, and release standards to mature end-user compute tool development

Lead Analytics Consultant (AVP)

Wells Fargo

T-shaped role providing leadership, oversight, and direct development in ETL, visualization, and advanced analytics workstreams.

Phoenix, Arizona

Jan 2018 - Jun 2022

- Onboarded applications to Pivotal Cloud Foundry with Splunk logging and Blackduck security scans
- Built web scrape automation for business intelligence visualization previews and validation
- Created dynamic databases for manual and automated data ingestion from flat-file data providers
- Engaged technology peer groups to execute high-priority regulatory data analytics

Earlier roles: Applications Systems Engineer (AVP), Business Systems Consultant (AVP), Technology Manager, Team Lead / Leadership Development Program / Operations

Wells Fargo / Bank of America

Phoenix, Arizona

1999 - Jan 2018

PROJECTS

vramfit

Hardware-targeted LLM quantization.

saucier

Provenance-first extraction: parses Escoffier's January 1909 edition into a catalogue of 151 preparations where every claim traces to a source line, with no model in the loop.

turboquant-vllm

First TurboQuant vLLM plugin, built paper-to-PyPI in 72 hours after Google's ICLR 2026 publication.

typevet

Asks an open model typed questions about text and images and reads each answer as a probability from its next-token distribution; runs Gemma 4 31B on a 24 GiB card (llama.cpp or vLLM).

judgevet

Typed Python client, CLI and MCP server for TypeSafe's Jev judgment model, with a published calibration study: Jev's scam-text calibration gap measured at 7 points (ECE 0.073) with sorting score 0.995.

EDUCATION

Bachelor of Science, Accounting Information Systems
DeVry University

AWARDS & HONORS

Named as co-inventor on a second and third pending patent application 2026

Named as co-inventor on a pending patent application January 2026

Top Performer 2018 (New Orleans) - Recognition for vulnerability remediation efforts

Top Performer 2014 (Nashville) - Recognition for asset reporting initiatives

SKILLS

Core Competencies: Generative AI & Agents (Google ADK, Pydantic-AI), Python, Cloud Engineering (GCP, OpenShift/Kubernetes, PCF), Streaming & Observability (Kafka, Grafana), Multi-agent engineering workflows (specifier to builder to independent reviewer), Data Engineering & Pipelines, OCR & Document Intelligence, Video Inference, CI/CD & DevOps, Enterprise Architecture

Languages: English (native), Spanish (bilingual)